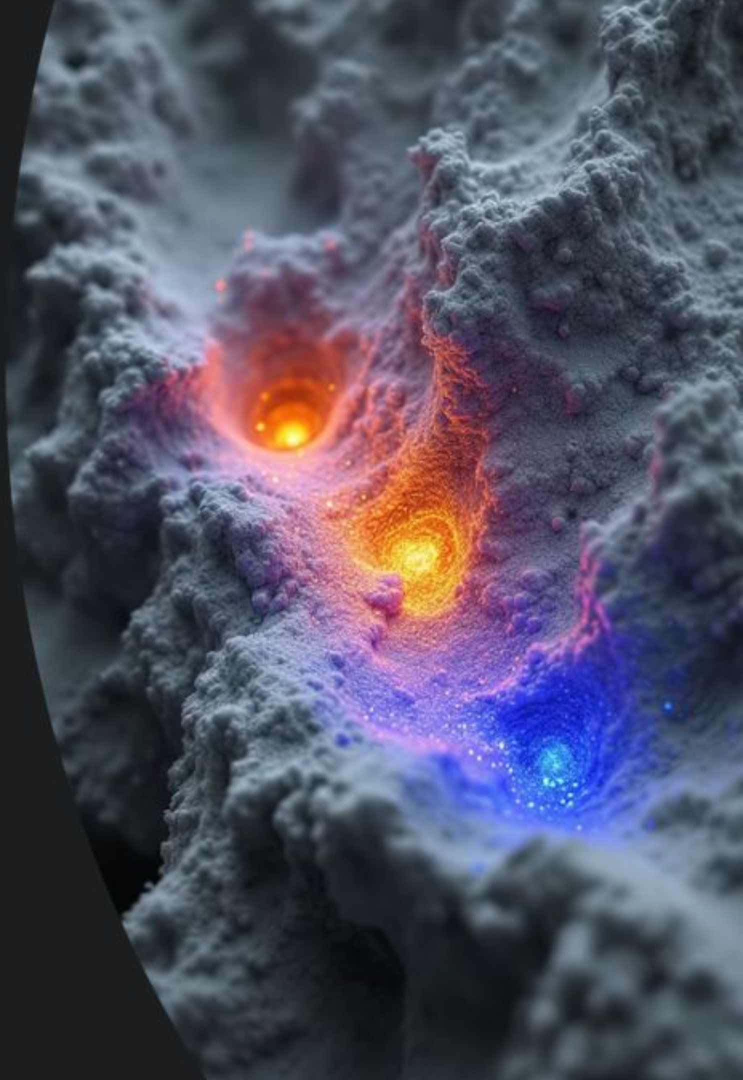
Uni-3DAD: Universal 3D Anomaly Detection

A unified framework for detecting all types of defects in manufactured items during food production

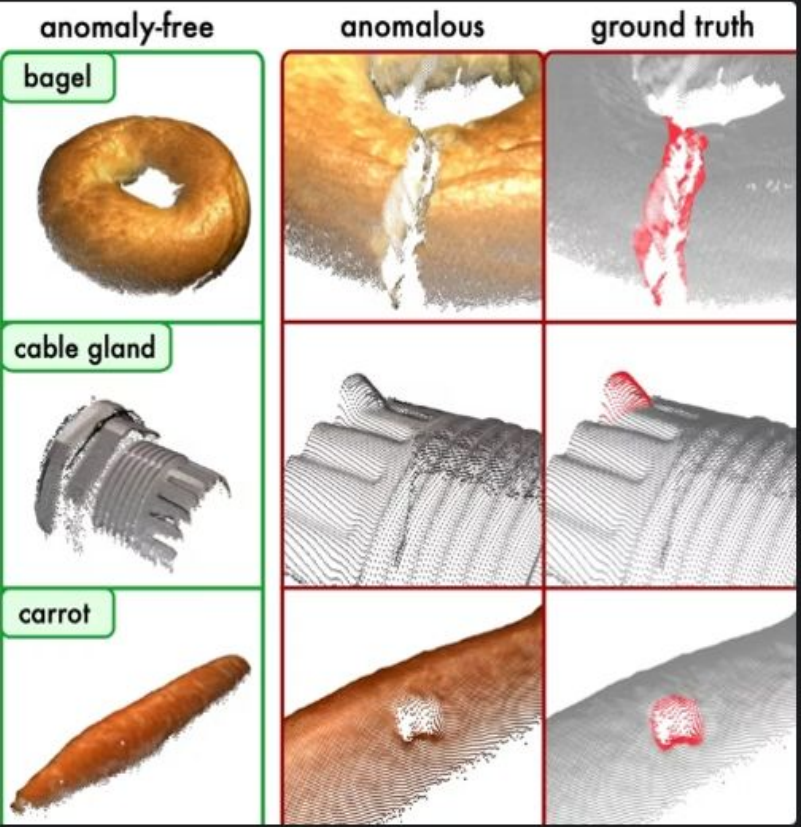
by Aaryan Sonawane, Trevor Tisdale & Aditya Dhodapkar



Why Anomaly Detection?

- Detecting faults in factory items is crucial for quality control
- Historically this has been done by humans with the naked eye, but computers are better equipped for this job because they don't lose focus and can spot defects the naked eye cannot see
- Most parts on a manufacturing line have a CAD file to refer to as a blueprint. Our challenge was to spot these defects in food items where there is no CAD file to refer to

Methods



- 3D point clouds are used to map the objects digitally onto a computer. This ensures to get a full 3D map of the object and factors such as a lighting or camera angles don't affect how the shape of the object is perceived like they are in a photograph

- In objects containing defects, point cloud data helps identify tiny cracks

- However large missing chunks in objects are hard to detect because there are no neighboring dots when the missing chunks are large

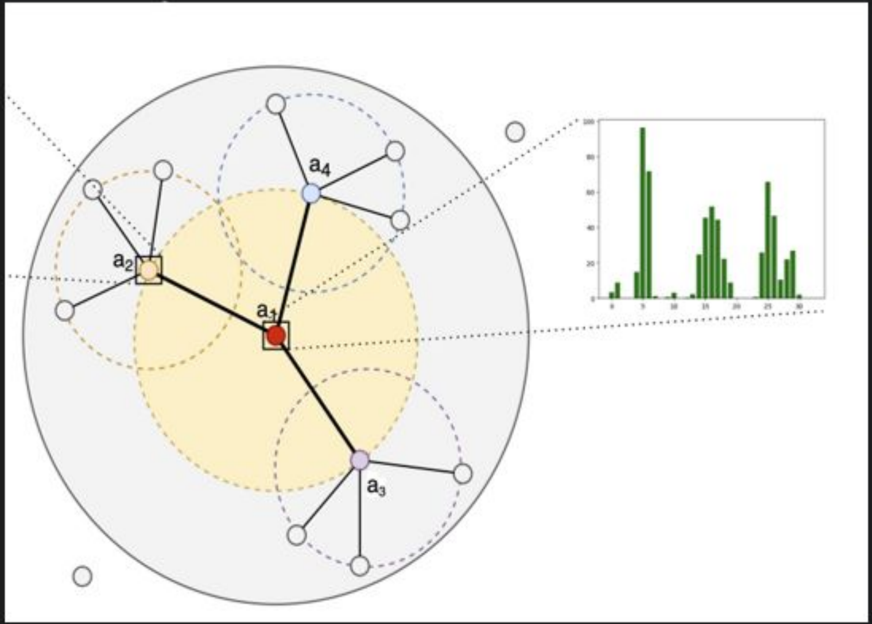
Uni3DAD Solution

- To combat the large missing chunks in the object, a combination of the point cloud method along with a generative method are used. This is where the complete Uni3DAD solution consisting of two parts comes into play to find all kinds of defects

  - Feature based point cloud checks for tiny surface defects and cracks
  - Reconstruction based detection which generates the entire image of a given object to act as a blueprint to detect large anomalies

- These two methods are combined for a singular anomaly detection solution called One-class Support Vector Machine (OCSVM) to fuse the detection results from both methods

Mathematical Approach

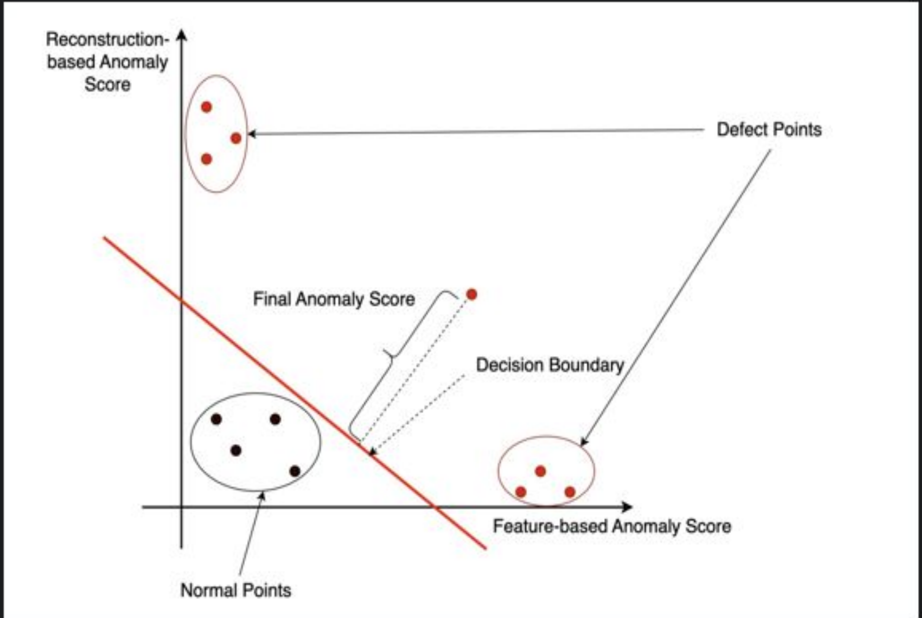


- Point clouds are created using 3D scanners to create a digital map of the given object
  - A point cloud is represented by a set of points $P = \{p_1, p_2, \ldots . p_n\}$ where each point $p_i = \{x_i, y_i, z_i\}$ is a dot in 3D space.

- For the feature-based detection for each point p_i the computer looks at 30 neighboring points to determine the angles in all 3 axes using the k nearest neighbor (KNN) algorithm between the points to determine the shape of the object.

  - These angles for each point p_i are displayed on a Fast Point Feature Histogram (FPFH)
  - The distance and angle values are stored in a vector f_i. An object with an anomaly will have a different value for the vector f_i than a standard object. This difference is given as s_1

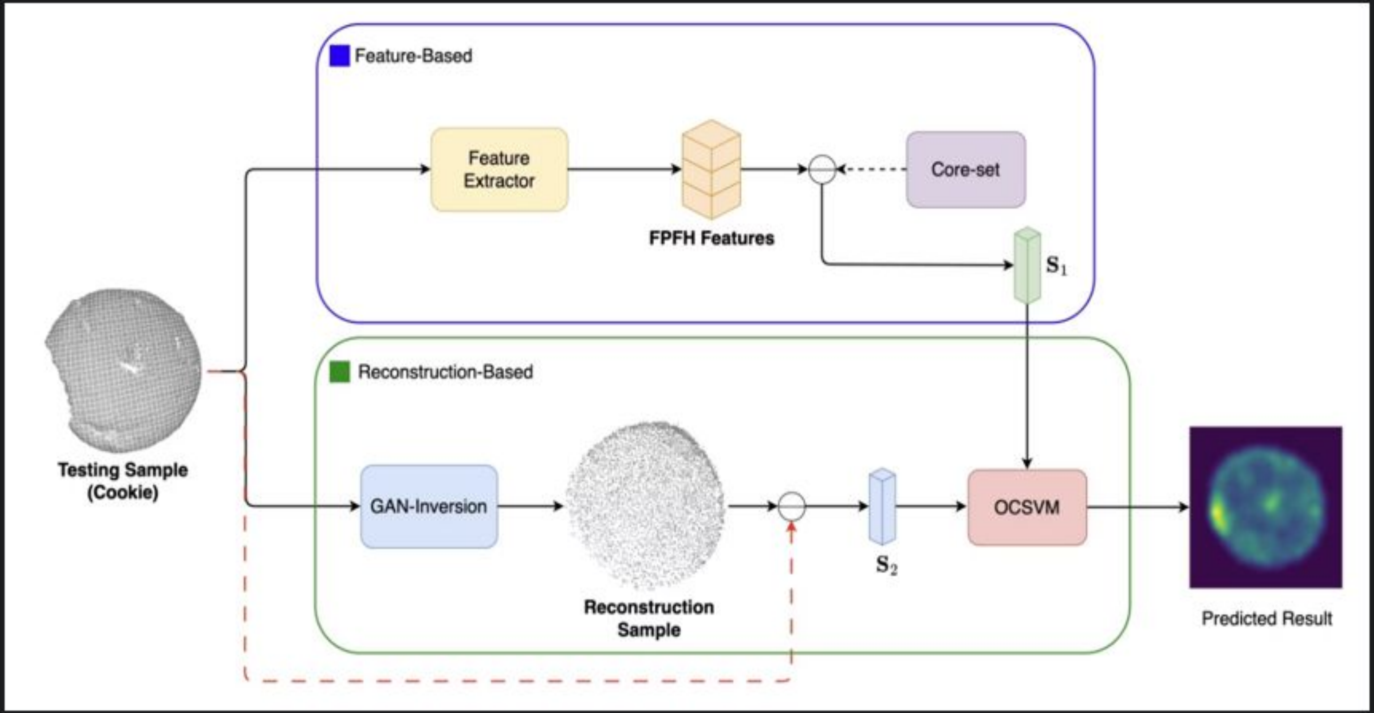
Mathematical Approach



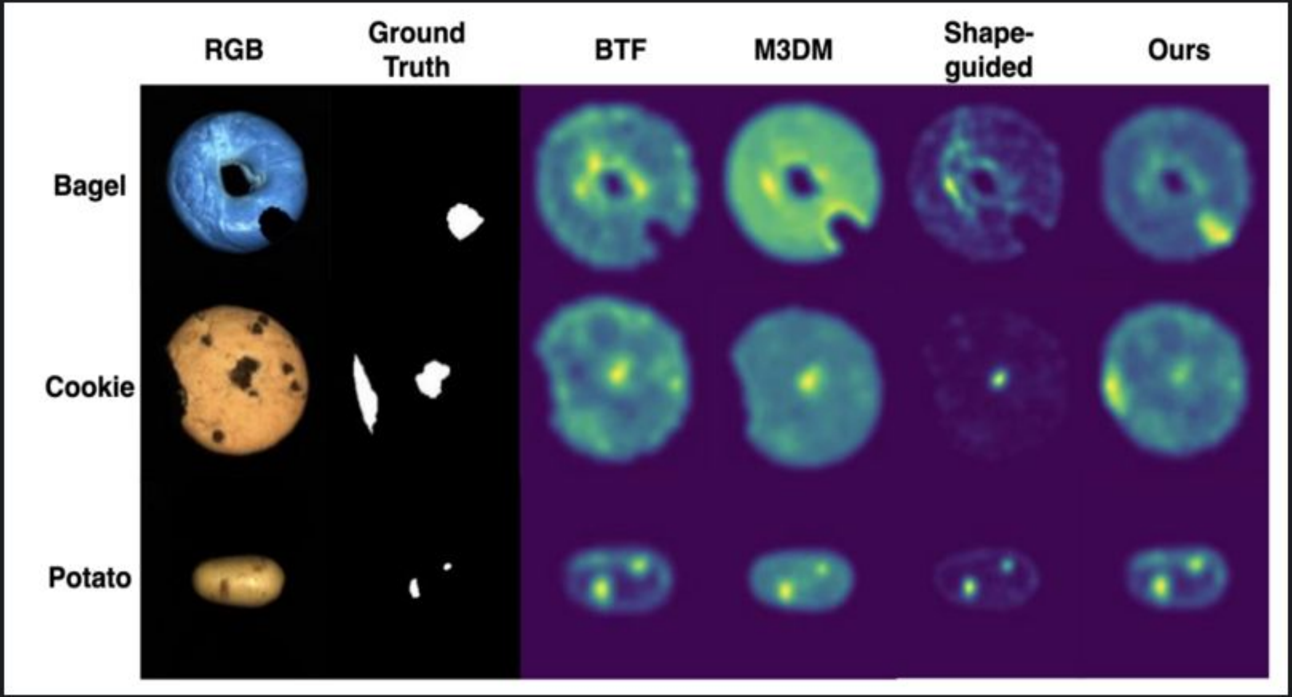
- For the reconstruction-based detection, a generative adversarial network (GAN) model is used. The GAN model is a generative AI method used to generate full images of what a given object is supposed to look like. This model is trained using MVTEC 3D-AD dataset. The computer records this difference score as s_2

- These scores are combined and plotted on 2D axes with s_1 on the x axis and s_2 on the y axis. This process is called one class support vector machine (OCSVM). Based on a cutoff slope on the plot, the given images are either deemed as normal, feature based anomalies or reconstruction-based anomalies.

Mathematical Approach



Experimental Results



Our method outperforms current approaches on incomplete shapes